

Identifying Validator Alliances by Voting Similarity in PoS Blockchain Governance via Visual Analytics System

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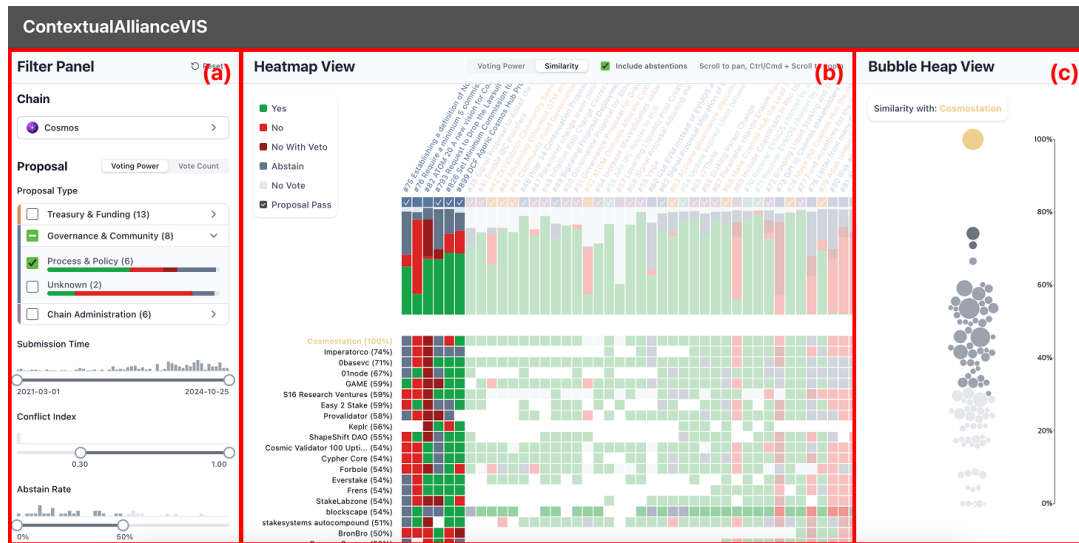


Figure 1: ContextualAllianceVIS(<https://contextualalliancevis.vercel.app>). (a) Filtering panel to select chain, proposals, and validators. (b) Heatmap for exploring voting patterns and alliances. (c) Bubble heap for examining similarity and influence of validators.

ABSTRACT

We present a visual analytics system to increase proposal approval likelihood in Proof-of-Stake(PoS) blockchain governance. The proposed system introduces the *Contextual Alliance Index(CAI)*, a similarity metric reflecting contextual information based on proposal voting data. Through heatmaps and bubble heap graphs, users can explore alliances among validators and support comparison and prioritization of persuadable validators. Furthermore, a case study illustrates an analytical process for understanding voting patterns for each proposal, and for identifying alliances and the prioritization of persuadable validators when drafting new proposals. This study is expected to contribute to the in-depth analysis of proposal patterns and the development of effective proposal strategies by identifying validator alliances in PoS blockchain governance.

Index Terms: Voting Pattern, Alliance Analysis, Blockchain Visualization, Similarity-Based Analysis

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1 INTRODUCTION

In a PoS blockchain network, validators verify blocks and transactions based on the amount of staked tokens, and governance decisions are made through proposals and voting [3]. When submitting a new proposal, it is important to identify validators who are likely to support the proposing validator and who can be persuaded in order to increase the likelihood of proposal approval. However, existing governance data analysis and visualization techniques primarily focus on visualizing voting power and simple voting results, making it difficult to capture both voting similarity and validator influence intuitively. Moreover, there is a lack of interactive tools that enable efficient exploration of alliance relationships [1]. Therefore, this study proposes an analysis system that calculates the voting similarity between validators based on proposal data from the Cosmos ecosystem, combines a heatmap that shows voting opinion and patterns of validators and a bubble heap graph [2] that shows both voting similarity and voting power among validators. This system supports the exploration of complex governance patterns by chain and proposal type, as well as prioritizing persuadable validators when drafting new proposals.

2 METHODOLOGY

This study analyzes 3,594 proposals and voting data from 3,421 validators collected from 17 chains within the Cosmos ecosystem up to November 2024. Each proposal includes five types of voting choices—‘Yes’, ‘No’, ‘Veto’, ‘Abstain’, and ‘Did not vote’—and voting power information. Using this data, the system calculates the similarity of voting behavior among validators and visualizes alliance structures from the perspective of the proposing validator.

2.1 Requirements

The system is designed to meet the following requirements: **(R1) Contextual and pattern-based analysis of proposals:** The system should enable users to filter proposals with significant opinion divergence and understand the voting patterns and contextual characteristics of each proposal. **(R2) Exploration of similarity and opposition among validators:** Given a specific validator, the system should enable quick identification of groups with similar or opposing voting behaviors. **(R3) Identification of persuadable validators based on voting power:** Among validators with similar voting opinions, the system should support the prioritization of persuadable validators based on each validator's voting power.

2.2 Metrics & Data Processing

To quantify the similarity between validators, this study introduces the *Contextual Alliance Index (CAI)*, which incorporates contextual weights into vote alignment scores. The similarity between two validators, A and B, is defined as follows:

$$\text{Contextual Alliance Index}_{A,B} = \alpha \cdot \frac{1}{N} \sum_{i=1}^N A_i \cdot W_i \quad (1)$$

Here, A_i represents the **Partial Alignment Score**, which measures the degree of agreement between the two validators on proposal i , with scores assigned based on full agreement, partial agreement, or disagreement. W_i is the **Contextual Weight** for proposal i , calculated as the arithmetic mean of the following three components: the **Conflict Weight** C_i [4], based on the level of opinion divergence; the **Minority Weight** M_i , indicating whether the vote reflects a minority opinion; and the **Recency Weight** R_i , reflecting proposal recency. The scaling factor α normalizes the overall score to ensure that the CAI does not exceed 100%, thereby preventing any single proposal from having disproportionate influence. Based on the computed CAI, validators are classified into three groups with respect to a given validator: **Aligned** ($CAI \geq 70\%$), **Neutral** ($30\% \leq CAI < 70\%$), and **Divergent** ($CAI < 30\%$). This classification enables clear identification of allies and persuadable validators.

For interpretation of voting patterns and context, proposal types were redefined through a BERT-based NLP pipeline followed by refinement using a LLM. The result is a hierarchical classification consisting of four high-level types and sixteen detailed subtypes. Detailed CAI formulas and definitions of proposal types are provided in Appendix Sections 1 and 2.

3 SYSTEM INTERFACE & INTERACTION

The system provides a web-based interface composed of a filtering panel and two visualization views. This allows users to visually explore voting patterns, similarity, and influence by selecting blockchains, proposals, and validators for analysis.

First, the **filtering panel** (Figure 1a) enables users to select chains, proposals, and validators according to contextual needs (R1). In the **heatmap view** (Figure 1b), validators' voting patterns for each proposal are visualized. Under the proposal title, the proposal type and pass/fail status are indicated through type-specific colors and a check icon, respectively. The stacked bar chart summarizes validators' voting results for each proposal and allows users to compare voting power and vote counts interactively. The heatmap displays each validator's vote for every proposal (R2). In the **bubble heap view** (Figure 1c), users can instantly grasp the distribution and influence of similar validators based on the size, density, and position of the bubbles (R3). We have provided a detailed description of the system interface and interactions in Section 3 of the appendix.

4 USAGE SCENARIO

The validator *Cosmostation*, active on the *Cosmos* chain, intends to submit a new proposal classified under the *Process & Policy* type.

Figure 1 illustrates the system output for this scenario. To determine the prioritization of persuadable validators based on past voting data, *Cosmos* chain and the *Process & Policy* type, which belongs to the broader *Governance & Community* type, are selected. To avoid proposals with extreme polarization, the Conflict Index is adjusted to range from 0.3 to 1. Proposals with an abstention rate exceeding 50% are excluded. For validators, only those who make up 99% of the total voting power and have participated in at least 40% of proposals are included, in order to focus on those with practical influence in governance decisions.

A total of 91 validators and 6 proposals were selected through filtering. Among these, 4 proposals were approved and 2 were rejected. The rejected proposals included one that required a minimum 5% fee from all validators and another that redefined the vision of the Cosmos Hub. The approved proposals involved setting the definition of NoWithVeto, the withdrawal of a lawsuit, setting a 5% minimum fee, and allocating community funds. The analysis reveals that *Cosmostation* tends to oppose fee-related proposals, abstain from those concerning rule definitions or legal matters, and support proposals related to community funding (R1).

Next, to identify similar validators, selecting *Cosmostation* in the heatmap automatically sorts the remaining validators based on their CAI scores. Validators such as *Imperatorco*, *Obasevc*, and *01node* appear at the top of the heatmap, indicating that their voting patterns are similar to that of *Cosmostation* (R2).

In the bubble heap on the right, two validators are identified as belonging to the Aligned group based on their color, and validators starting from *01node*, which has a 67% similarity score, fall into the Neutral group. To determine the prioritization of persuadable validators, the Neutral group is examined in descending order of bubble size, revealing *Everstake*, *Allnodes*, *Game*, and *Informal Systems* as the top candidates (R3).

5 CONCLUSION

In this study, we propose a system that enables the simultaneous exploration of voting similarity and influence of validators. The system utilizes a heatmap sorted by the calculated CAI and a bubble heap to effectively analyze governance patterns and persuasion priorities across different chains and proposal types. This study contributes a quantitative visual analysis approach that overcomes the limitations of simple voting results and similarity-based visualizations, enabling the simultaneous identification of alliance and influence structures based on context.

In the future, the system may be further enhanced by (1) integrating multi-chain activity data to analyze alliances from multiple perspectives, and (2) addressing the limitations of group classifications that rely solely on similarity without capturing contextual information, for example by allowing groups to be dynamically adjusted according to context.

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